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|-------------------------|-------------------------------------------------|
| <b>EXPT NO: 6</b>       | <b>SIFT AND HOG FEATURES FOR IMAGE ANALYSIS</b> |
| <b>DATE: 06/09/2025</b> |                                                 |

### AIM:

To perform the Utilization of SIFT and HOG features for image analysis.

### ALGORITHM:

1. Read image and convert to grayscale.
2. Initialize **SIFT/HOG detector** (cv2.SIFT\_create(), cv2.HOGDescriptor()).
3. Detect keypoints and **compute descriptors**.
4. Draw **keypoints** on the image.
5. Display or store the **extracted features**.
6. Use features for **comparison or matching**.

### CODE:

```
import cv2

import matplotlib.pyplot as plt
from skimage.feature import hog
from skimage import color

# ----- SIFT -----
img = cv2.imread('/content/drive/MyDrive/input.jpg')

# Check if image was loaded successfully
if img is None:
    print("Error: Could not load image from the specified path. Please check
the file path and ensure the image exists.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # SIFT detector
```

```

sift = cv2.SIFT_create()
keypoints, descriptors = sift.detectAndCompute(gray, None)

# Draw keypoints
sift_img = cv2.drawKeypoints(img, keypoints, None,
flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS)

# ----- HOG -----
# Convert to grayscale for HOG
gray_hog = color.rgb2gray(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))

# Compute HOG features and visualization
hog_features, hog_img = hog(gray_hog, orientations=9, pixels_per_cell=(8,
8),
                           cells_per_block=(2, 2), visualize=True,
channel_axis=None)

# ----- Display Results -----
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(sift_img, cv2.COLOR_BGR2RGB))
plt.title("SIFT Features")
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(hog_img, cmap='gray')
plt.title("HOG Features")
plt.axis('off')

plt.show()

```

## OUTPUT:

SIFT Features



HOG Features



## RESULT:

Thus, Utilization of SIFT and HOG features for image analysis was implemented successfully.